

In Sect. 5.1 we considered binary regression models, that is, regression situations where the response is observed in two categories. In many applications, from social science to medicine, response variables often have more than two categories. For example, consumers may choose between different brands of a product or they may express their opinion about some product in ordered categories ranging from “very satisfied” to “not satisfied at all.” Similarly, voters choose between several parties or they assess the quality of candidates in ordered categories. In medicine, we may, for example, not only distinguish between “infection” and “no infection” but also between several types of infection, as in Example 6.1 below. Another application with a categorical response is Example 1.4 (p. 9) on forest health, where the status of trees is assessed in ordered categories, such as “no damage”, “medium damage”, and “severe damage.” This chapter extends regression models and methods for binary responses to the case of categorical responses with more than two categories. Compared to binary regression there is a greater variety of models, depending on the type of response and (underlying) response mechanisms. In particular, we distinguish between models for responses with unordered categories (Sect. 6.2) and ordered categories (Sect. 6.3).

6.1 Introduction

In many applications, the response variable is not binary as in Sect. 5.1, but rather multi-categorical. Such a response variable can be defined as either *ordinal*, i.e., the categories $1, \dots, c + 1$ of the response can be ordered, or *nominal*, which means that the categories $1, \dots, c + 1$ are unordered. We distinguish $c + 1$ categories as we will later define c dummy variables representing the response variable and choose the $(c + 1)$ th category as the reference.

Example 6.1 Caesarian Delivery—Categorical Response

Table 6.1 shows data on infections after C-sections performed at a clinical center in Munich; see also Example 5.3 (p. 284) of Chap. 5. The response variable has three unordered

Table 6.1 Data on infections for 251 C-sections

	C-section					
	Planned			Unplanned		
	Infection			Infection		
	I	II	no	I	II	no
Antibiotics						
Risk factor	0	1	17	4	7	87
No risk factor	0	0	2	0	0	0
No antibiotics						
Risk factor	11	17	30	10	13	3
No risk factor	4	4	32	0	0	9

Table 6.2 Pulmonary function test

Age	Smoking status	Breathing test		
		Normal	Borderline	Abnormal
<40	Nonsmoker	577	27	7
	Former smoker	192	20	3
	Current smoker	682	46	11
40–59	Nonsmoker	164	4	0
	Former smoker	145	15	7
	Current smoker	245	47	27

categories: type I infection, type II infection, and no infection. In addition, information on three covariates is available:

- NPLAN* : C-section was not planned/planned.
- RISK* : Presence/absence of risk factors, such as diabetes or obesity.
- ANTIB* : Antibiotics were prescribed/not prescribed as prophylaxis.
- △

Example 6.2 Forest Health Status—Categorical Response

Example 1.4 (p. 9) studies the health status of beech trees measured in terms of the response variable “degree of defoliation” with nine ordered categories (0 %, 12.5 %, . . . , 100 % defoliation). We aggregate the response variable into three categories, 1 = no (0 %), 2 = weak (12.5 %–37.5 %), and 3 = severe (≥ 50 %) defoliation, since some original response categories have only a few observations or are completely missing. For every observation in this study, the continuous and categorical covariates that are listed in Table 1.5 (p. 10) have been collected annually along with the response variable. In contrast to Example 6.1, this data set contains ungrouped, individual observations.

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Example 6.3 Pulmonary Function—Categorical Response

In a study on the impairment of pulmonary function, the age and the smoking behavior of Texan industrial workers were collected as covariates; see [Forthofer and Lehnén \(1981\)](#). The results of a breathing test with categories “normal,” “borderline,” and “abnormal” are considered as the dependent variable. The resulting data are summarized in the contingency table given in Table 6.2.

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In all these examples and for the remainder of this chapter, the response variable Y has $c + 1$ ordered or unordered categories, i.e., $Y \in \{1, \dots, c + 1\}$. Categorical regression models relate the probabilities

$$\pi_r = P(Y = r), \quad r = 1, \dots, c + 1,$$

to covariates. To formulate such models, it is useful to represent the response variable by a vector $\mathbf{y} = (y_1, \dots, y_c)'$ of c dummy variables

$$y_r = \begin{cases} 1, & Y = r \\ 0, & \text{otherwise} \end{cases} \quad r = 1, \dots, c.$$

Here, we have chosen category $c + 1$ as the “reference” category so that $y_{c+1} = 1 - y_1 - \dots - y_c$. We therefore obtain

$$Y = r \iff \mathbf{y} = (0, \dots, 1, \dots, 0)', \quad r = 1, \dots, c,$$

with a value of 1 as the r th component of $\mathbf{y}$, and

$$\pi_r = P(Y = r) = P(y_r = 1), \quad r = 1, \dots, c.$$

Correspondingly, for the reference category, we have

$$Y = c + 1 \iff \mathbf{y} = (0, \dots, 0, \dots, 0)', \quad P(Y = c + 1) = 1 - \pi_1 - \dots - \pi_c.$$

Multinomial Distribution

The multinomial distribution provides the generalization of the binomial distribution that is necessary for modeling categorical responses. For an individual observation $\mathbf{y} = (y_1, \dots, y_c)'$ with binary 0/1 variables $y_1, \dots, y_c$ and vector $\boldsymbol{\pi} = (\pi_1, \dots, \pi_c)'$ of occurrence probabilities, the probability function is given by

$$f(\mathbf{y} | \boldsymbol{\pi}) = \pi_1^{y_1} \cdot \dots \cdot \pi_c^{y_c} (1 - \pi_1 - \dots - \pi_c)^{1 - y_1 - \dots - y_c}.$$

For m independent trials, y_r represents the number of repetitions of category r , $r = 1, \dots, c$. The multinomial probability function of $\mathbf{y} = (y_1, \dots, y_c)'$ is then given by

$$f(\mathbf{y} | \boldsymbol{\pi}) = \frac{m!}{y_1! \cdot \dots \cdot y_c! (m - y_1 - \dots - y_c)!} \pi_1^{y_1} \cdot \dots \cdot \pi_c^{y_c} (1 - \pi_1 - \dots - \pi_c)^{1 - y_1 - \dots - y_c}. \quad (6.1)$$

We write

$$\mathbf{y} \sim M(m, \boldsymbol{\pi}),$$

with the parameters m (number of trials) and $\boldsymbol{\pi} = (\pi_1, \dots, \pi_c)'$ (occurrence probabilities). The first two moments of the multinomial distribution are given by

$$E(\mathbf{y}) = m\boldsymbol{\pi} = \begin{pmatrix} m\pi_1 \\ \vdots \\ m\pi_c \end{pmatrix}, \quad \text{Cov}(\mathbf{y}) = m \begin{pmatrix} \pi_1(1 - \pi_1) & \cdots & -\pi_1\pi_c \\ \vdots & \ddots & \vdots \\ -\pi_c\pi_1 & \cdots & \pi_c(1 - \pi_c) \end{pmatrix}.$$

When using the relative frequencies $\bar{y}_r = y_r/m$ instead of the absolute frequencies y_r , the relative frequency vector $\bar{\mathbf{y}} = (\bar{y}_1, \dots, \bar{y}_c)'$ follows a scaled multinomial distribution, $\bar{\mathbf{y}} \sim M(m, \boldsymbol{\pi})/m$, such that

$$E(\bar{\mathbf{y}}) = \begin{pmatrix} \pi_1 \\ \vdots \\ \pi_c \end{pmatrix}, \quad \text{Cov}(\bar{\mathbf{y}}) = \frac{1}{m} \begin{pmatrix} \pi_1(1 - \pi_1) & \cdots & -\pi_1\pi_c \\ \vdots & \ddots & \vdots \\ -\pi_c\pi_1 & \cdots & \pi_c(1 - \pi_c) \end{pmatrix}.$$

Data

The data structure is analogous to binary and other univariate generalized linear models. For ungrouped individual data, the value of the categorical response variable $Y_i \in \{1, \dots, c + 1\}$ or $\mathbf{y}_i = (y_{i1}, \dots, y_{ic})'$ and the covariate vector $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})'$ for each unit i are given:

$$\begin{array}{l} \text{Unit 1} \\ \vdots \\ \text{Unit } i \\ \vdots \\ \text{Unit } n \end{array} \begin{pmatrix} \mathbf{y}'_1 = (y_{11}, \dots, y_{1c}) \\ \vdots \\ \mathbf{y}'_i = (y_{i1}, \dots, y_{ic}) \\ \vdots \\ \mathbf{y}'_n = (y_{n1}, \dots, y_{nc}) \end{pmatrix} \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{i1} & \cdots & x_{ik} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}.$$

With grouped data, we combine observations with identical covariate vectors $\mathbf{x}_i$ into a group i of n_i units. Then, absolute frequencies $\mathbf{y}_i = (y_{i1}, \dots, y_{ic})'$ or relative frequencies $\bar{\mathbf{y}}_i = (\bar{y}_{i1}, \dots, \bar{y}_{ic})'$ of observed categories in group i , $i = 1, \dots, G$ are collected along with the covariates:

$$\begin{array}{l} \text{Group 1} \\ \vdots \\ \text{Group } i \\ \vdots \\ \text{Group } G \end{array} \begin{pmatrix} n_1 \\ \vdots \\ n_i \\ \vdots \\ n_G \end{pmatrix} \begin{pmatrix} \bar{\mathbf{y}}'_1 = (\bar{y}_{11}, \dots, \bar{y}_{1c}) \\ \vdots \\ \bar{\mathbf{y}}'_i = (\bar{y}_{i1}, \dots, \bar{y}_{ic}) \\ \vdots \\ \bar{\mathbf{y}}'_G = (\bar{y}_{G1}, \dots, \bar{y}_{Gc}) \end{pmatrix} \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{i1} & \cdots & x_{ik} \\ \vdots & \vdots & & \vdots \\ 1 & x_{G1} & \cdots & x_{Gk} \end{pmatrix}.$$

Grouped data are often presented in contingency tables, as in Examples 6.1 and 6.3, indicating absolute frequencies $y_{i1}, \dots, y_{ic}$ instead of relative frequencies.

Given the covariates $\mathbf{x}_i$, individual data $\mathbf{y}_i$ follow a multinomial distribution with

$$\mathbf{y}_i \sim \mathbf{M}(1, \boldsymbol{\pi}_i), \quad \boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{ic})'.$$

For grouped data, the absolute frequencies are multinomial distributed with

$$\mathbf{y}_i \sim \mathbf{M}(n_i, \boldsymbol{\pi}_i),$$

whereas the relative frequencies $\bar{\mathbf{y}}_i$ follow a scaled multinomial distribution. Categorical regression aims at modeling and estimating the probabilities $\pi_{ir} = \mathbf{P}(Y_i = r) = \mathbf{P}(y_{ir} = 1)$ depending on covariates $\mathbf{x}_i$.

In comparison to binary regression models, multinomial regression has a larger variety of modeling possibilities. In the next two sections, we describe some important models for nominal and ordinal response variables. Additional models can be found in [Agresti \(2002\)](#), [Fahrmeir and Tutz \(2001\)](#), and [Tutz \(2011\)](#).

6.2 Models for Unordered Categories

In this section, we assume that the categories $r \in \{1, \dots, c + 1\}$ either do not have an ordered structure or this structure is not modeled in the analyses, which is the case for [Example 6.1](#) and for many applications in biomedicine, social, and economic sciences. Well-known areas of application are, for example, the modeling of decisions, the choice of political party, or the mode of transportation when going to work (e.g., car, train, bike). The categories of the response variable simply represent the different alternatives without requiring any ordered structure. This implies that the response variable is of a nominal scale.

For binary response variables ($c = 1$), we considered the logit model

$$\mathbf{P}(Y_i = 1) = \pi_i = \frac{\exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik})}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik})} = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})},$$

as presented in [Sect. 5.1](#). A direct generalization is the categorical (multinomial) logit model defined in [Box 6.1](#).

It follows from [Eq. \(6.3\)](#) that the linear predictor $\eta_{ir} = \mathbf{x}_i' \boldsymbol{\beta}_r$ specifies the (logarithmic) *odds* or *relative risk* between category r and the reference category $c + 1$ in terms of either a log-linear or exponentially multiplicative model. Thus, the interpretation of parameters is analogous to the binary logit model. The only difference is that we do not model the logit for presence ($Y = 1$) in relation to absence ($Y = 0$) but rather the different logits for category r ($Y = r$) relative to the reference category $c + 1$ ($Y = c + 1$).

When interpreting the covariate effects, we have to be aware that a positive regression coefficient β_{rj} does not necessarily imply an increasing probability for category r as x_j increases. It does hold that the odds for category r increase relative

6.1 Multinomial Logit Model

Data

The categorical response variable $Y_i \in \{1, \dots, c + 1\}$ is measured on a nominal scale. In addition, covariates $\mathbf{x}_i$, which are independent of the response category, are given.

Model

The probability of occurrence for category r is specified as

$$P(Y_i = r) = \pi_{ir} = \frac{\exp(\mathbf{x}'_i \boldsymbol{\beta}_r)}{1 + \sum_{s=1}^c \exp(\mathbf{x}'_i \boldsymbol{\beta}_s)} \quad r = 1, \dots, c. \quad (6.2)$$

For the reference category, we have

$$\pi_{i,c+1} = 1 - \pi_{i1} - \dots - \pi_{ic} = \frac{1}{1 + \sum_{s=1}^c \exp(\mathbf{x}'_i \boldsymbol{\beta}_s)}.$$

An equivalent representation is

$$\log \frac{\pi_{ir}}{\pi_{i,c+1}} = \mathbf{x}'_i \boldsymbol{\beta}_r \quad \text{or} \quad \frac{\pi_{ir}}{\pi_{i,c+1}} = \exp(\mathbf{x}'_i \boldsymbol{\beta}_r), \quad r = 1, \dots, c. \quad (6.3)$$

The parameter vectors $\boldsymbol{\beta}_r = (\beta_{r0}, \beta_{r1}, \dots, \beta_{rk})'$ and therefore the linear predictor $\eta_{ir} = \mathbf{x}'_i \boldsymbol{\beta}_r = \beta_{r0} + x_{i1}\beta_{r1} + \dots + x_{ik}\beta_{rk}$ are specific for each category, $r = 1, \dots, c$.

to the reference category. However, this does not imply that the odds for category r relative to all other alternatives also increase. If, for example, category s has a higher regression coefficient than category r , i.e., $\beta_{sj} > \beta_{rj}$, the odds for category s relative to the reference category grows faster than that of category r . Hence, it is possible that the probability for category r decreases despite its positive coefficient β_{rj} . To assess how a changing covariate affects probabilities, it is useful to visualize the probabilities in a plot with all of the remaining covariates kept fixed, e.g., at average values. The most straightforward interpretation is, however, in terms of the log-odds or odds (6.3).

Table 6.3 C-section births: grouped data

		Response		Covariates		
		y_1	y_2	$NPLAN$	$RISK$	$ANTIB$
Group 1	$n_1 = 40$	4	4	0	0	0
Group 2	$n_2 = 58$	11	17	0	1	0
Group 3	$n_3 = 2$	0	0	0	0	1
Group 4	$n_4 = 18$	0	1	0	1	1
Group 5	$n_5 = 9$	0	0	1	0	0
Group 6	$n_6 = 26$	10	13	1	1	0
Group 7	$n_7 = 98$	4	7	1	1	1

Example 6.4 Caesarian Delivery—Categorical Response

The response variable Y “infection” has three categories: “infection type I” ($Y = 1$), “infection type II” ($Y = 2$), and “no infection” ($Y = 3$). With $Y = 3$ as reference category, we code the response variable through $\mathbf{y}' = (y_1, y_2)$, with

$$\begin{aligned} \mathbf{y}' &= (1, 0) && \text{infection of type I} \\ \mathbf{y}' &= (0, 1) && \text{infection of type II} \\ \mathbf{y}' &= (0, 0) && \text{no infection} \end{aligned}$$

Similar to Example 5.2 (p. 278), we use the binary covariates $NPLAN$, $RISK$, and $ANTIB$. The grouped data arising from the contingency table of Example 6.1 can be found in Table 6.3.

We consider a main effects multinomial logit model based on (6.3), i.e.,

$$\log \frac{P(\text{infection type } r)}{P(\text{no infection})} = \beta_{r0} + \beta_{rN}NPLAN + \beta_{rR}RISK + \beta_{rA}ANTIB, \quad r = 1, 2,$$

or equivalently

$$\frac{P(\text{infection type } r)}{P(\text{no infection})} = \exp(\beta_{r0}) \exp(\beta_{rN}NPLAN) \exp(\beta_{rR}RISK) \exp(\beta_{rA}ANTIB).$$

As such, $\exp(\beta_{rN})$ is the multiplicative effect of the binary covariate $NPLAN$ for category r , when the covariate has the value 1 instead of 0. Alternatively, we can also interpret $\exp(\beta_{rN})$ as the increase in relative risk (odds ratio) if the covariate takes a value 1 relative to 0. If $NPLAN = 0$ is increased to $NPLAN = 1$, we have

$$\exp(\beta_{rN}) = \frac{P(\text{type } r \mid NPLAN = 1, R, A)}{P(\text{no} \mid NPLAN = 1, R, A)} \bigg/ \frac{P(\text{type } r \mid NPLAN = 0, R, A)}{P(\text{no} \mid NPLAN = 0, R, A)}.$$

Analogous interpretations hold for the covariates $RISK$ and $ANTIB$. Table 6.4 provides the estimated parameters for the given data. The estimates are based on the methods provided in Sect. 6.4.

According to these parameter estimates, antibiotics decrease the relative risk for both types of infections, with a somewhat stronger effect for the type I infections. We also find that the relative risk strongly increases when risk factors exist or when a C-section was not planned. We will analyze whether or not the differences in effects for type I and type II infections are significant in more detail in Sect. 6.4.

Table 6.4 C-section births: estimated coefficients

Type I infection			Type II infection		
	β	$\exp(\beta)$		β	$\exp(\beta)$
<i>intercept</i>	-2.621	0.072	<i>intercept</i>	-2.560	0.077
<i>NPLAN</i>	1.174	3.235	<i>NPLAN</i>	0.996	2.707
<i>ANTIB</i>	-3.520	0.030	<i>ANTIB</i>	-3.087	0.046
<i>RISK</i>	1.829	6.228	<i>RISK</i>	2.195	8.980

The results are based on the STATA function `mlogit`

Table 6.5 C-section births: estimated probabilities for “no infection,” infection type I, and infection type II

<i>NPLAN</i>	<i>ANTIB</i>	<i>RISK</i>	P(no infection)	P(type I)	P(type II)
0	0	0	0.870	0.063	0.067
1	0	0	0.692	0.163	0.145
0	1	0	0.994	0.002	0.004
0	0	1	0.466	0.211	0.323
1	1	0	0.984	0.007	0.009
1	0	1	0.230	0.337	0.433
0	1	1	0.957	0.013	0.030
1	1	1	0.886	0.038	0.076

Since the estimated parameters cannot easily be transferred to the underlying probabilities, Table 6.5 provides the estimated probabilities for all possible covariate combinations. We see that infections of type I or II are generally unlikely. Exceptions are the cases where risk factors are present *and* antibiotics are not given (*RISK* = 1 and *ANTIB* = 0). Then the probability of “no infection” is below 50%; 0.466 if C-section was planned (*NPLAN* = 0) and 0.230 if C-section was not planned (*NPLAN* = 1). On the other hand, the probabilities for type I and II infections are rather high in these cases.

△

Nominal Models and Latent Utility Models

The multinomial logit model, as well as other categorical regression models, especially probit models, can be motivated and derived with the help of latent utility models. This is not only interesting for alternative interpretations but also useful for statistical inference, especially for Bayesian approaches; see also Sect. 5.6.3.

We assume that each of the $c + 1$ alternatives is associated with a specific utility and that the random utility function of the r th alternative has the form

$$u_r = \tilde{\eta}_r + \varepsilon_r,$$

where $\tilde{\eta}_r$ is a constant (the utility of the r th alternative), and ε_r is a random variable with cumulative distribution function F . According to the principle of maximum utility, we observe alternative r if the associated (unobserved) utility is maximal, i.e.,

$$Y = r \iff u_r = \max_{s=1, \dots, c+1} u_s.$$

Assuming independent errors ε_r with extreme maximal-value distribution $F(x) = \exp(-\exp(-x))$ yields

$$P(Y = r) = \frac{\exp(\tilde{\eta}_r)}{\sum_{s=1}^{c+1} \exp(\tilde{\eta}_s)}, \quad r = 1, \dots, c+1;$$

see [McFadden \(1973\)](#). Since only the difference in utility functions is identifiable, we choose a reference category. With category $c+1$ as the reference category, we obtain

$$P(Y = r) = \frac{\exp(\tilde{\eta}_r - \tilde{\eta}_{c+1})}{1 + \sum_{s=1}^c \exp(\tilde{\eta}_s - \tilde{\eta}_{c+1})} = \frac{\exp(\eta_r)}{1 + \sum_{s=1}^c \exp(\eta_s)},$$

with $\eta_r = \tilde{\eta}_r - \tilde{\eta}_{c+1}$.

An extension of this concept results when the utilities do not only depend on the covariates characterizing the statistical units facing the decision but also on the alternatives. Let $\mathbf{x}_i$ denote the covariates that have already been included in η_{ir} and $\mathbf{w}_{ir}$ the vector of covariates that are specific to the alternative r . If the mode of transportation is considered as the decision problem, the cost or time associated with a specific mode of transportation provides examples of *category-specific covariates*. Since category-specific covariates often also vary between the individual observations, we use a double subscript on $\mathbf{w}_{ir}$, where i represents the individual, while r represents the alternative. Assuming independent errors that follow an extreme maximal-value distribution leads to the extended multinomial logit model

$$\pi_{ir} = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta}_r + (\mathbf{w}_{ir} - \mathbf{w}_{i,c+1})' \boldsymbol{\gamma})}{1 + \sum_{s=1}^c \exp(\mathbf{x}_i' \boldsymbol{\beta}_s + (\mathbf{w}_{is} - \mathbf{w}_{i,c+1})' \boldsymbol{\gamma})},$$

$r = 1, \dots, c$. The model has *category-unspecific (global)* covariates $\mathbf{x}_i$, which are supplemented with *category-specific* coefficients $\boldsymbol{\beta}_r$, and *category-specific* covariates $\mathbf{w}_{ir} - \mathbf{w}_{i,c+1}$, which are accompanied by *category-unspecific (global)* coefficients $\boldsymbol{\gamma}$. The difference from $\mathbf{w}_{ir}$ to $\mathbf{w}_{i,c+1}$ of the reference category results from building the difference $\eta_{ir} = \tilde{\eta}_{ir} - \tilde{\eta}_{i,c+1} = \mathbf{w}_{ir}' \boldsymbol{\gamma} - \mathbf{w}_{i,c+1}' \boldsymbol{\gamma}$.

Other distributional assumptions for the errors in the latent utility model lead to alternative models. If we assume independent but standard normal distributed errors ε_r , we obtain the (“independent”) probit model. If we choose a multivariate normal distribution (with a nondiagonal covariance matrix) for the errors $\boldsymbol{\varepsilon}$, we obtain the multivariate probit model.

6.3 Ordinal Models

In this section, we present models that are useful when the response variable is ordinal, i.e., the categories can be ordered. The explicit use of the ordinal scale of Y allows for models with parsimonious parameterization. Ordinal responses have been present in the examples on forest health (Example 6.2), as well as in the example regarding pulmonary function (Example 6.3).

6.3.1 The Cumulative Model

The most widely used ordinal regression model is motivated from a threshold mechanism, similar to the binary regression model in Sect. 5.1 (p. 274). We can derive the model by assuming a latent (unobserved) variable that drives the decision for the observed alternatives. Exceeding a certain threshold on the latent scale results in an observable category of the response variable.

Let u denote the unobserved latent variable. In the examples on forest health or the pulmonary function test, u can be considered the latent damage state of a tree or the lung. The latent variable corresponding to covariate value $\mathbf{x}_i$ (excluding the intercept) is assumed to be

$$u_i = -\mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \quad (6.4)$$

where $\boldsymbol{\beta}$ is the parameter vector and ε_i is an error variable with cumulative distribution function F . The reason for the minus sign in front of $\mathbf{x}_i' \boldsymbol{\beta}$ will become clear later. The link between observed and latent variable is defined by the threshold mechanism

$$Y_i = r \iff \theta_{r-1} < u_i \leq \theta_r, \quad r = 1, \dots, c+1, \quad (6.5)$$

where $-\infty = \theta_0 < \theta_1 < \dots < \theta_{c+1} = \infty$ are the latent, ordered thresholds placed on the latent continuum. Note that the intercept has to be excluded from the predictor $\mathbf{x}_i' \boldsymbol{\beta}$ in order to obtain an identifiable model. Otherwise, a shift of the intercept and a corresponding negative shift of the thresholds would lead to an equivalent model.

Category r is observed, if the latent variable falls between the thresholds θ_{r-1} and θ_r . Interpreted as a binary decision, we obtain a response variable $Y_i \leq r$, if $u_i \leq \theta_r$, i.e., when the latent variable remains under the threshold θ_r , and $Y_i > r$ if $u_i > \theta_r$.

Figure 6.1 illustrates the threshold concept. It shows the density of the latent variable $u_i = -\mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$ for two different realizations of $\mathbf{x}_i$. The corresponding densities do have the same form ($\varepsilon_i \sim F$), but they are shifted on the latent continuum. The area between two thresholds corresponds to the probability of a certain category as exemplified by the grey area in Fig. 6.1 for the second category. Obviously there is a strong change in the probability due to the shift

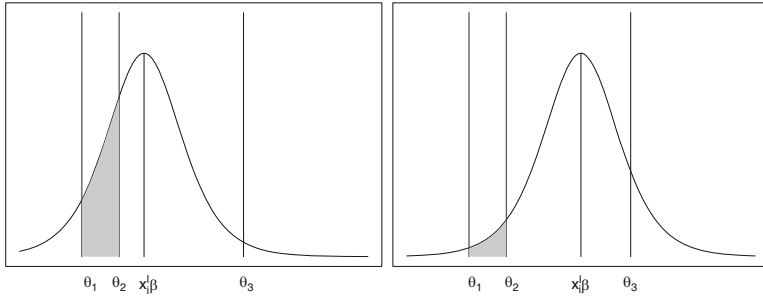


Fig. 6.1 Thresholds and densities of the latent variable

forced through x_i . The illustration in Fig. 6.1 is based on a logistic density, but all statements remain unchanged when using a different distribution for the error terms.

The *cumulative model* with distribution function F results from the two assumptions (6.4) and (6.5) through

$$\begin{aligned}
 P(Y_i \leq r) &= P(u_i \leq \theta_r) \\
 &= P(-x_i' \beta + \varepsilon_i \leq \theta_r) \\
 &= P(\varepsilon_i \leq \theta_r + x_i' \beta) \\
 &= F(\theta_r + x_i' \beta), \quad r = 1, \dots, c + 1.
 \end{aligned}$$

The name “cumulative model” refers to the specification of cumulative probabilities $P(Y_i \leq r) = P(Y_i = 1) + \dots + P(Y_i = r)$ on the left-hand side of the model equation. The model itself no longer contains the latent variable and can be considered a regression model with regressors x_i and parameters $\theta_1, \dots, \theta_c$ and β . The occurrence probabilities are given by

$$P(Y_i = r) = F(\theta_r + x_i' \beta) - F(\theta_{r-1} + x_i' \beta), \quad r = 1, \dots, c + 1.$$

Depending on the choice of F , we obtain different models. For example, the *cumulative logit model* results for the logistic distribution function F with

$$P(Y_i \leq r) = \frac{\exp(\theta_r + x_i' \beta)}{1 + \exp(\theta_r + x_i' \beta)}$$

or equivalently

$$\log \frac{P(Y_i \leq r)}{P(Y_i > r)} = \theta_r + x_i' \beta.$$

6.2 Cumulative Model

Data

The response variable $Y_i \in \{1, \dots, c+1\}$ is categorical and measured on an ordinal scale. The covariates $\mathbf{x}_i$ do not depend on the response categories.

Model

The response variable Y_i is linked to a latent variable u_i via

$$Y_i = r \iff \theta_{r-1} < u_i \leq \theta_r, \quad r = 1, \dots, c+1,$$

with thresholds $-\infty = \theta_0 < \theta_1 < \dots < \theta_{c+1} = \infty$. A linear model

$$u_i = -\mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i,$$

with error distribution F is assumed for u_i . The *cumulative model* for Y_i is then given by

$$P(Y_i \leq r) = F(\theta_r + \mathbf{x}_i' \boldsymbol{\beta}), \quad r = 1, \dots, c+1,$$

or equivalently

$$P(Y_i = r) = F(\theta_r + \mathbf{x}_i' \boldsymbol{\beta}) - F(\theta_{r-1} + \mathbf{x}_i' \boldsymbol{\beta}).$$

This model is also called the *proportional odds* model. The term *proportional* refers to the fact that the ratio of the cumulative odds for subpopulations characterized by $\mathbf{x}_i$ and $\tilde{\mathbf{x}}_i$ is given by

$$\frac{P(Y_i \leq r | \mathbf{x}_i) / P(Y_i > r | \mathbf{x}_i)}{P(Y_i \leq r | \tilde{\mathbf{x}}_i) / P(Y_i > r | \tilde{\mathbf{x}}_i)} = \exp((\mathbf{x}_i - \tilde{\mathbf{x}}_i)' \boldsymbol{\beta}). \quad (6.6)$$

The ratio is independent of category r , and therefore the cumulative odds are proportional across all categories.

If we choose F to be the extreme minimum-value distribution $F(x) = 1 - \exp(-\exp(x))$, we obtain the *cumulative extreme value* or *grouped Cox model*

$$P(Y_i \leq r) = 1 - \exp(-\exp(\theta_r + \mathbf{x}_i' \boldsymbol{\beta})), \quad (6.7)$$

or

$$\log(-\log P(Y_i > r)) = \theta_r + \mathbf{x}_i' \boldsymbol{\beta}.$$

The term “grouped Cox” or “proportional hazards model” stems from a property analogous to Eq. (6.6) of an underlying duration time model. With the reparameterization $\tilde{\theta}_r = \log(\exp(\theta_r) - \exp(\theta_{r-1}))$, $r = 1, \dots, c$, we can equivalently express Eq. (6.7) as

$$P(Y = r \mid Y \geq r) = 1 - \exp(-\exp(\tilde{\theta}_r + \mathbf{x}'_i \boldsymbol{\beta})) = F(\tilde{\theta}_r + \mathbf{x}'_i \boldsymbol{\beta}). \quad (6.8)$$

On the left-hand side, we find the probability of the response variable in category r , given that the category r or a larger category is reached. In duration time analysis (when Y represents a discrete time), this conditional probability is referred to as the discrete hazard function. Model (6.8) is the discrete-time analogue of the continuous-time Cox model from duration time analysis.

Example 6.5 Forest Health Status—Cumulative Logit Model

We model the three-categorical response variable representing the health status with a cumulative logit model. The effects of *year* and *age* are approximated by orthogonal cubic polynomials (see Example 3.5 on p. 90), whereas all other continuous covariates from Table 1.5 of Example 1.4 are modeled with linear effects. All binary or categorical covariates are effect-coded with the most common category as reference category (coded as -1). In this analysis, we do not consider the spatial information, which is provided by the location of a tree. The results in Table 6.6 and Fig. 6.2 should therefore be mainly considered as illustrations of a preliminary nature. Estimation is based on the methods outlined in the next section. In Sect. 9.5, we will further analyze the data with a more flexible geoadditive model. The nonlinear effects of age and year shown in Fig. 6.2 are only very rough approximations for the corresponding covariate effects as we will see in Sect. 9.5. There is evidence that younger trees do generally show a lower damage state than middle-aged and older trees. The calendar effect *year* reflects the observed temporal trends. The risk of damage increases starting in 1983, flattens towards the end of the decade and then shows a continuous decrease from 1990 to 2000. In recent years, however, we again find some evidence of an increasing risk. To get an intuitive feel about the variability of the probabilities for the three damage states it is recommended to plot the estimated probabilities in addition to the effect plots. As an example, Fig. 6.3 visualizes the probabilities for “no damage,” “weak damage,” and “severe damage” depending on the age of the trees. Thereby we kept the effects of the other continuous covariates fixed at their mean value and of the categorical covariates at their mode category; see also Examples 2.8 (p. 35) and 2.13 (p. 54) of Chap. 2.

We only comment on some selected linear effects: A high canopy density, the presence of fertilization, and a moderately dry soil significantly reduce the risk of damage. In contrast, humid soil and a homogeneous deciduous forest increase the probability of damage. We reanalyze the data in Example 9.7 (p. 556) using a flexible geoadditive model, which shows that some effects react sensitively to model choice, especially when spatial effects are included.

△



6.3.2 The Sequential Model

An alternative to the cumulative model is the sequential model, which also explicitly uses the ordinal scaling of the response categories. Due to its construction, the sequential model is useful when the categories of the response variable are obtained

Table 6.6 Forest health status: estimated covariate effects. The results are obtained using function `ologit` of the software STATA

Variable	Coefficient	Standard error	t-statistic	p-value	95 % Confidence interval	
θ_0	-5.8994	1.298			-8.4444	-3.3544
θ_1	-1.8424	1.290			-4.3709	0.6861
<i>gradient</i>	0.0004	0.007	0.05	0.959	-0.0130	0.0137
<i>canopyd</i>	-0.0319	0.003	-10.51	<0.001	-0.0378	-0.0259
<i>alt</i>	-0.0004	0.001	-0.26	0.794	-0.0030	0.0023
<i>depth</i>	-0.0206	0.007	-2.98	0.003	-0.0341	-0.0070
<i>ph</i>	-0.8598	0.250	-3.44	0.001	-1.3501	-0.3695
<i>type</i>	-0.3963	0.074	5.35	<0.001	0.2512	0.5414
<i>fert</i>	-0.4453	0.125	-3.56	<0.001	-0.6902	-0.2004
<i>humus0</i>	-0.1256	0.132	-0.95	0.343	-0.3851	0.1340
<i>humus2</i>	0.3100	0.116	2.67	0.007	0.0828	0.5373
<i>humus3</i>	0.1688	0.134	1.26	0.209	-0.0943	0.4319
<i>humus4</i>	-0.2739	0.179	-1.53	0.126	-0.6247	0.0769
<i>watermoisture1</i>	-0.6270	0.150	-4.17	<0.001	-0.9217	-0.3323
<i>watermoisture3</i>	0.4926	0.108	4.58	<0.001	0.2817	0.7036
<i>alkali1</i>	0.3646	0.159	2.30	0.022	0.0536	0.6756
<i>alkali3</i>	-0.5451	0.156	-3.50	<0.001	-0.8500	-0.2401
<i>alkali4</i>	0.9230	0.221	4.18	<0.001	0.4901	1.3559

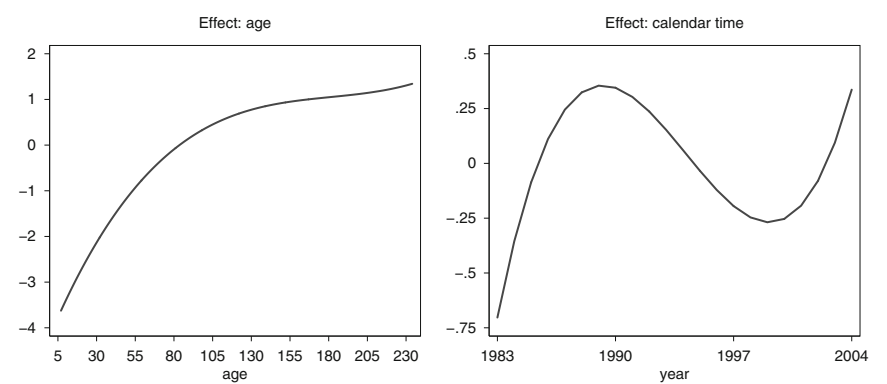


Fig. 6.2 Forest health status: estimated polynomial effects of age and calendar time

sequentially. If we assume that, in Example 6.3, the pulmonary function is originally normal, the abnormal category (for given smoking status and age) can only be reached provided that the borderline category has been reached in between. Another example would be years of unemployment considered as categories of a duration time response. One can only be unemployed for two years provided that one has been unemployed for one year before. The assumptions on which we build

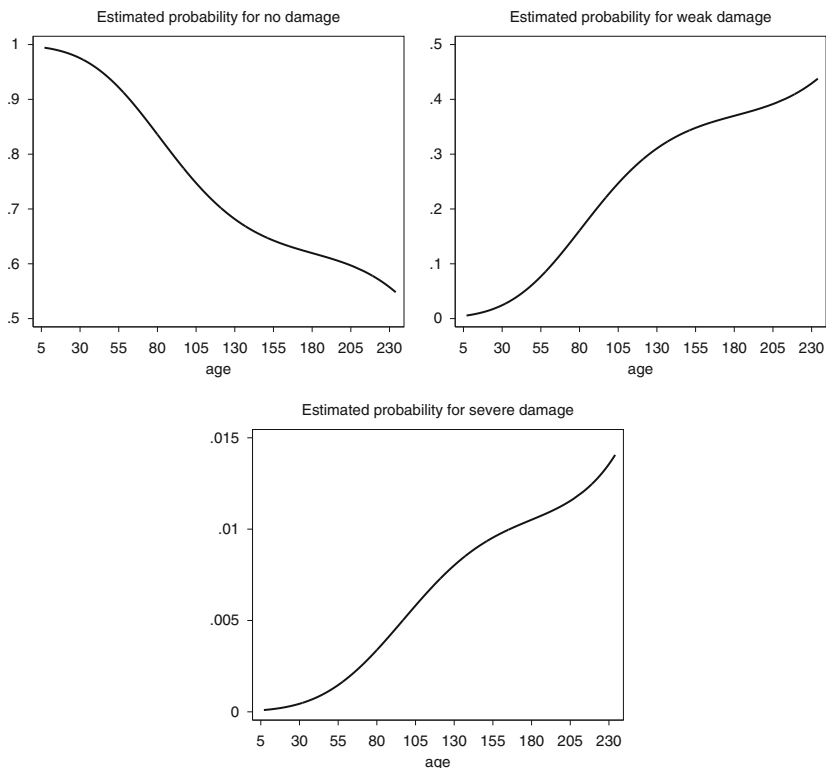


Fig. 6.3 Forest health status: estimated probabilities for “no damage,” “weak damage,” and “severe damage” depending on age. The remaining continuous covariates are held fixed at their mean values. The categorical covariates are held fixed at their mode category

sequential models are, thus, successively or gradually achievable categories of the variable $Y \in \{1, \dots, c + 1\}$.

The sequential reachability of the categories is modeled explicitly in terms of a sequence of binary transitions. The process starts in $Y_i = 1$ and a binary model of Sect. 5.1 describes the transition to $Y_i > 1$:

$$P(Y_i = 1) = F(\theta_1 + \mathbf{x}_i' \boldsymbol{\beta}).$$

If the response variable remains in category 1, the process stops. Otherwise, the following transition is dichotomously modeled with

$$P(Y_i = 2 | Y_i \geq 2) = F(\theta_2 + \mathbf{x}_i' \boldsymbol{\beta})$$

as the conditional probability for remaining in $Y_i = 2$ and complementary probability $1 - P(Y_i = 2 | Y_i \geq 2)$ for the transition to $Y_i > 2$.

Accordingly, the r th step of the sequential mechanism is based on

$$P(Y_i = r \mid Y_i \geq r) = F(\theta_r + \mathbf{x}'_i \boldsymbol{\beta}), \quad r = 1, \dots, c. \quad (6.9)$$

The process stops as soon as one of the binary transition equations decides for the process to remain in category r .

It should be pointed out that we are not actually observing individual transitions but only the realized category of the response variable. The sequential mechanism is basically an assumption which helps to formulate the sequential model. Alternatively, model (6.9) can be expressed in terms of marginal probabilities

$$P(Y_i = r) = F(\theta_r + \mathbf{x}'_i \boldsymbol{\beta}) \prod_{s=1}^{r-1} (1 - F(\theta_s + \mathbf{x}'_i \boldsymbol{\beta})), \quad r = 1, \dots, c,$$

and

$$P(Y_i = c + 1) = 1 - \sum_{s=1}^c P(Y_i = s)$$

for the reference category.

For the logistic distribution function F , we specifically obtain the *sequential logistic model*; for the extreme minimum-value distribution $F(x) = 1 - \exp(-\exp(x))$, we obtain the *extreme value sequential model*. Possible alternatives are the probit model or other binary regression models. The sequential logit model can be represented through the conditional transition probabilities

$$\log \frac{P(Y_i = r \mid Y_i \geq r)}{1 - P(Y_i = r \mid Y_i \geq r)} = \theta_r + \mathbf{x}'_i \boldsymbol{\beta}.$$

When the number of categories is small, an extended sequential model with category-specific regression coefficients for some of the covariates can be considered. Instead of the linear term $\theta_r + \mathbf{x}'_i \boldsymbol{\beta}$ in Eq. (6.9), we obtain the general linear predictor

$$\theta_r + \mathbf{x}'_i \boldsymbol{\beta}_r + \mathbf{z}'_i \boldsymbol{\gamma},$$

where $\mathbf{x}_i$ and $\mathbf{z}_i$ represent groups of variables with category-specific effects $\boldsymbol{\beta}_r$ and global effects $\boldsymbol{\gamma}$, respectively.

Example 6.6 Pulmonary Function—Sequential Model

We specify the influence of age and smoking behavior on the test results of Table 6.2 using a sequential model. Dummy coding is used for the explanatory variables with reference categories $age \geq 40$ and smoking status “current smoker.” The results provided in Table 6.7 are based on `remlreg` objects of `BayesX` and are generally as expected. Younger workers of age lower than 40 and nonsmokers are less likely to show a borderline or abnormal breathing test compared to older workers of age larger than 40 and current smokers. The difference between former and current smokers is comparably small (and nonsignificant).



6.3 Sequential Models

Data

The response variable $Y_i \in \{1, \dots, c + 1\}$ is categorical and measured on an ordinal scale. The categories can only be reached sequentially.

Model

The transitions between the categories are modeled using binary regression models. If category $r \in \{1, \dots, c\}$ has already been reached, the (potential) termination of the process in r is defined by

$$P(Y_i = r \mid Y_i \geq r) = F(\theta_r + \mathbf{x}'_i \boldsymbol{\beta}).$$

Thus, we have

$$P(Y_i = r) = F(\theta_r + \mathbf{x}'_i \boldsymbol{\beta}) \prod_{s=1}^{r-1} (1 - F(\theta_s + \mathbf{x}'_i \boldsymbol{\beta})), \quad r = 1, \dots, c.$$

If F is the logistic distribution function, we obtain the sequential logit model. If F is the standard normal distribution function, the sequential probit model results. We can also consider category-specific effects of covariates based on the general predictor $\theta_r + \mathbf{x}'_i \boldsymbol{\beta}_r + \mathbf{z}'_i \boldsymbol{\gamma}$.

Table 6.7 Lung function test: estimated effects for the main effects sequential model

Variable	Coefficient	Standard error	p-value	95 % Confidence interval	
θ_1	1.495	0.120	<0.001	1.260	1.730
θ_2	0.551	0.179	0.002	0.200	0.901
<i>age</i> < 40	−0.742	0.135	<0.001	−1.007	−0.478
<i>non sm.</i>	−0.867	0.175	<0.001	−1.210	0.522
<i>former sm.</i>	−0.188	0.171	0.272	−0.523	0.147

We now additionally include interactions between age and smoking status; see Table 6.8 for the results. The interaction model has a comparably lower AIC of 1575.13 compared to the main effects model which shows an AIC of 1596.59. Quite surprisingly (at least at first sight), the interaction effects are both positive. For instance, nonsmoking workers with age below 40 show a considerably high interaction effect of 2.220 indicating that the risk of borderline and abnormal breathing test is increased. However, it is not justified to interpret the main effects and the interaction effects separately. It is rather necessary to compute the effects of the six combinations of age and smoking behavior. For instance, for the combination *age* < 40 and “nonsmoking,” we obtain the effect

$$-1.207 \cdot 1 - 2.519 \cdot 1 - 0.586 \cdot 0 + 2.220 \cdot 1 + 0.832 \cdot 0 = -1.506.$$

Table 6.8 Lung function test: estimated effects for the sequential model with interactions

Variable	Coefficient	Standard error	p-value	95 % Confidence interval	
θ_1	1.249	0.125	<0.001	1.004	1.495
θ_2	0.387	0.181	0.032	0.032	0.742
<i>age</i> < 40	-1.207	0.174	<0.001	-1.548	-0.865
<i>non sm.</i>	-2.519	0.521	<0.001	-3.541	-1.496
<i>former sm.</i>	-0.586	0.234	0.012	-1.045	-0.127
<i>age</i> < 40, <i>non</i>	2.220	0.560	<0.001	-1.122	3.318
<i>age</i> < 40, <i>former</i>	0.832	0.339	0.014	0.167	1.498

Table 6.9 Lung function test: combined effects in the main effects plus interactions model

Covariate combination	Effect
<i>age</i> < 40, <i>nonsmoker</i>	$-1.207 - 2.519 + 2.220 = -1.506$
<i>age</i> < 40, <i>former smoker</i>	$-1.207 - 0.586 + 0.832 = -0.961$
<i>age</i> < 40, <i>current smoker</i>	-1.207
<i>age</i> $\geq$ 40, <i>nonsmoker</i>	-2.519
<i>age</i> $\geq$ 40, <i>former smoker</i>	-0.586
<i>age</i> $\geq$ 40, <i>current smoker</i>	0

Table 6.10 Lung function test: estimated probabilities for the sequential model with interactions

Covariate combination	P(normal)	P(borderline)	P(abnormal)
<i>age</i> < 40, <i>nonsmoker</i>	0.940	0.052	0.008
<i>age</i> < 40, <i>former smoker</i>	0.901	0.078	0.020
<i>age</i> < 40, <i>current smoker</i>	0.921	0.066	0.013
<i>age</i> $\geq$ 40, <i>nonsmoker</i>	0.977	0.021	0.001
<i>age</i> $\geq$ 40, <i>former smoker</i>	0.862	0.100	0.038
<i>age</i> $\geq$ 40, <i>current smoker</i>	0.778	0.133	0.090

Table 6.9 lists the calculated effects of all covariate combinations and Table 6.10 lists the estimated probabilities. We now clearly see that in both age groups the risk of borderline or abnormal breathing test is lower for nonsmokers and to a lesser extent for former smokers compared to current smokers. Surprisingly, the non-smokers in the age group *age* $\geq$ 40 have an even lower effect than in the group *age* < 40.

The example shows that it can be tedious to interpret models with interaction effects. We have already pointed out in Sect. 3.4 that the interaction between covariates can also be modeled by defining a new variable whose categories consist of *all* possible combinations of age and smoking status. The result of this approach with reference category *age* $\geq$ 40, *current smoker* is given in Table 6.11. We now immediately see the effects of the covariates without any tedious calculations (the differences in the third digit are due to rounding errors).

△

Table 6.11 Lung function test: estimated effects for the sequential model with interactions and alternative dummy coding

Variable	Coefficient	Standard error	p-value	95 % Confidence interval	
θ_1	1.249	0.125	<0.001	1.004	1.495
θ_2	0.387	0.181	0.032	0.032	0.742
<i>age</i> < 40, <i>non</i>	−1.505	0.201	<0.001	−1.899	−1.111
<i>age</i> < 40, <i>former</i>	−0.960	0.241	<0.001	−1.433	−0.487
<i>age</i> < 40, <i>current</i>	−1.207	0.174	<0.001	−1.548	−0.865
<i>age</i> ≥ 40, <i>non</i>	−2.519	0.521	<0.001	−3.541	−1.496
<i>age</i> ≥ 40, <i>former</i>	−0.586	0.234	0.013	−1.046	−0.127

6.4 Estimation and Testing: Likelihood Inference

Parameter estimation relies on the same maximum likelihood principles that have been used for binary regression models and more generally with GLMs in Chap. 5. We assume the more general case of grouped data where the responses y_{ir} , $i = 1, \dots, G$, $r = 1, \dots, c$ represent the number of repetitions of category r out of n_i independent trials. Since the response variables $\mathbf{y}_i = (y_{i1}, \dots, y_{ic})'$ follow multinomial distributions

$$\mathbf{y}_i \sim M(n_i, \boldsymbol{\pi}_i)$$

and are assumed to be (conditionally) independent, the likelihood function is given by the product

$$L(\boldsymbol{\beta}) = \prod_{i=1}^G f(\mathbf{y}_i | \boldsymbol{\pi}_i)$$

of densities $f(\mathbf{y}_i | \boldsymbol{\pi}_i)$ of the form (6.1).

The vector of probabilities $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{ic})'$ depends on the vector $\boldsymbol{\beta}$ of all parameters through linear predictors with specific forms depending on the chosen model. Up to an additive constant, the log-likelihood is given by

$$l(\boldsymbol{\beta}) = \sum_{i=1}^G (y_{i1} \log \pi_{i1} + \dots + y_{ic} \log \pi_{ic} + y_{i,c+1} \log \pi_{i,c+1})$$

with $y_{i,c+1} = n_i - y_{i1} - \dots - y_{ic}$ and $\pi_{i,c+1} = 1 - \pi_{i1} - \dots - \pi_{ic}$. From this, we can derive the score function $\mathbf{s}(\boldsymbol{\beta}) = \partial l(\boldsymbol{\beta}) / \partial \boldsymbol{\beta}$ as the vector of first derivatives and the expected information matrix

$$\mathbf{F}(\boldsymbol{\beta}) = E(\mathbf{s}(\boldsymbol{\beta})\mathbf{s}'(\boldsymbol{\beta})).$$

The ML estimator is computed as the iterative solution of the ML equations

$$\mathbf{s}(\hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

The score function and the information matrix as well as the iterative computation of the ML estimator are multivariate extensions of the quantities derived for binary regression models and univariate GLMs in Chap. 5; see the following details.



To allow for a unified presentation of all categorical regression models, we introduce some common notation. In every categorical regression model, the occurrence probabilities $P(Y_i = r) = P(y_{ir} = 1) = \pi_{ir}$ for category r are connected to linear predictors η_{ir} , $r = 1, \dots, c$ via response functions h_r , i.e.,

$$\pi_{ir} = h_r(\eta_{i1}, \dots, \eta_{ic}), \quad r = 1, \dots, c.$$

The specification of the response functions and the linear predictors depends on the chosen categorical regression model.

For the multinomial logit model in Box 6.1, we have

$$\pi_{ir} = \frac{\exp(\eta_{ir})}{1 + \sum_{s=1}^c \exp(\eta_{is})} = h_r(\eta_{i1}, \dots, \eta_{ic}),$$

with $\eta_{ir} = \mathbf{x}'_i \boldsymbol{\beta}_r = \beta_{r0} + x_{i1}\beta_{r1} + \dots + x_{ik}\beta_{rk}$. In the extended form with category-specific explanatory variables $\mathbf{w}_{ir} - \mathbf{w}_{i,c+1}$ and global parameters $\boldsymbol{\gamma}$, the predictor is extended to

$$\eta_{ir} = \mathbf{x}'_i \boldsymbol{\beta}_r + (\mathbf{w}_{ir} - \mathbf{w}_{i,c+1})' \boldsymbol{\gamma}. \quad (6.10)$$

For the ordinal cumulative model in Box 6.2, we have $\pi_{i1} = F(\eta_{i1}) = h_1(\eta_{i1}, \dots, \eta_{ic})$ and

$$\pi_{ir} = F(\eta_{ir}) - F(\eta_{i,r-1}) = h_r(\eta_{i1}, \dots, \eta_{ic}), \quad r = 2, \dots, c,$$

with

$$\eta_{ir} = \theta_r + x_{i1}\beta_1 + \dots + x_{ik}\beta_k.$$

Combining the predictors into the vector

$$\boldsymbol{\eta}_i = (\eta_{i1}, \dots, \eta_{ic})',$$

and the parameters to the parameter vector $\boldsymbol{\beta}$, we can always obtain the form

$$\boldsymbol{\eta}_i = \mathbf{X}_i \boldsymbol{\beta},$$

with an appropriately defined design matrix $\mathbf{X}_i$.

For the multinomial logit model with the extended predictor (6.10), we have $\boldsymbol{\beta} = (\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_c, \boldsymbol{\gamma})'$ and

$$\mathbf{X}_i = \begin{pmatrix} \mathbf{x}'_i & \mathbf{w}'_{i1} - \mathbf{w}'_{i,c+1} \\ \mathbf{x}'_i & \mathbf{w}'_{i2} - \mathbf{w}'_{i,c+1} \\ & \ddots & \vdots \\ & \mathbf{x}'_i & \mathbf{w}'_{ic} - \mathbf{w}'_{i,c+1} \end{pmatrix}.$$

For the ordinal cumulative models, the vector of regression coefficients is $\boldsymbol{\beta} = (\theta_1, \dots, \theta_c, \beta_1, \dots, \beta_k)'$ and the design matrix has to be defined as

$$X_i = \begin{pmatrix} 1 & & & \mathbf{x}'_i \\ & 1 & & \mathbf{x}'_i \\ & & \ddots & \vdots \\ & & & 1 & \mathbf{x}'_i \end{pmatrix}.$$

If we additionally introduce the c -dimensional response function

$$\mathbf{h}(\boldsymbol{\eta}_i) = (h_1(\boldsymbol{\eta}_i), \dots, h_c(\boldsymbol{\eta}_i))',$$

we can write all categorical regression models for $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{ic})'$ in compact matrix notation as

$$\boldsymbol{\pi}_i = \mathbf{h}(\boldsymbol{\eta}_i), \quad \boldsymbol{\eta}_i = X_i \boldsymbol{\beta}.$$

The score function $\mathbf{s}(\boldsymbol{\beta}) = \partial l(\boldsymbol{\beta}) / \partial \boldsymbol{\beta}$ of the multinomial distribution does have a structure that is similar to the univariate GLM, presented in Chap. 5. We have

$$\mathbf{s}(\boldsymbol{\beta}) = \sum_{i=1}^G X_i' \mathbf{D}_i \boldsymbol{\Sigma}_i^{-1} (\mathbf{y}_i - n_i \boldsymbol{\pi}_i),$$

where $\mathbf{D}_i = \partial \mathbf{h}(\boldsymbol{\eta}_i) / \partial \boldsymbol{\eta}$ is the $c \times c$ -matrix of the partial derivatives, evaluated at $\boldsymbol{\eta}_i = X_i \boldsymbol{\beta}$, and $\boldsymbol{\Sigma}_i$ is the covariance matrix

$$\boldsymbol{\Sigma}_i = n_i \begin{pmatrix} \pi_{i1}(1 - \pi_{i1}) & -\pi_{i1}\pi_{i2} & \cdots & -\pi_{i1}\pi_{ic} \\ -\pi_{i2}\pi_{i1} & \pi_{i2}(1 - \pi_{i2}) & & \\ \vdots & & \ddots & \vdots \\ -\pi_{ic}\pi_{i1} & \cdots & & \pi_{ic}(1 - \pi_{ic}) \end{pmatrix}.$$

Introducing the weight matrix

$$\mathbf{W}_i = \mathbf{D}_i \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i',$$

(that depends on $\boldsymbol{\beta}$), we can write the Fisher matrix as

$$\mathbf{F}(\boldsymbol{\beta}) = \sum_{i=1}^G X_i' \mathbf{W}_i X_i,$$

which again is the generalization of the Fisher matrix in univariate GLMs.

Combining all observations and probabilities in $\mathbf{y} = (\mathbf{y}'_1, \dots, \mathbf{y}'_G)'$ and $\boldsymbol{\pi} = (\boldsymbol{\pi}'_1, \dots, \boldsymbol{\pi}'_G)'$, defining the full design matrix

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_G \end{pmatrix}$$

for all observations, and forming the block-diagonal matrices $\Sigma = \text{blockdiag}(\Sigma_1, \dots, \Sigma_G)$, $D = \text{blockdiag}(D_1, \dots, D_G)$, and $W = \text{blockdiag}(W_1, \dots, W_G)$ finally yields

$$s(\beta) = X'D\Sigma^{-1}(y - n_i\pi), \quad F(\beta) = X'WX.$$

Numerical Computation of the ML Estimator

We can again use the Fisher scoring method discussed in Sect. 5.1.2 for the iterative solution of the ML equations with $(t + 1)$ -th iteration given by

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + F^{-1}(\hat{\beta}^{(t)})s(\hat{\beta}^{(t)}).$$

The iterations can also be expressed as iteratively (re)weighted least squares updates

$$\hat{\beta}^{(t+1)} = (X'W(\hat{\beta}^{(t)})X)^{-1}X'W(\hat{\beta}^{(t)})\tilde{y}^{(t)},$$

based on the working observations $\tilde{y}^{(t)} = (\tilde{y}_1(\hat{\beta}^{(t)}), \dots, \tilde{y}_G(\hat{\beta}^{(t)}))$, with components

$$\tilde{y}_i(\hat{\beta}^{(t)}) = X_i\hat{\beta}^{(t)} + (D_i^{-1}(\hat{\beta}^{(t)}))'(y_i - \pi_i(\hat{\beta}^{(t)})).$$



Asymptotic Properties and Tests of Linear Hypotheses

The asymptotic properties of the ML estimator are analogous to those in the univariate case. Under relatively weak regularity conditions and with increasing sample size $n \rightarrow \infty$, the ML estimator $\hat{\beta}$ exists, is consistent, and is asymptotically normal, i.e.,

$$\hat{\beta} \overset{a}{\sim} N(\beta, F^{-1}(\hat{\beta}));$$

see Fahrmeir and Kaufmann (1985). Tests of linear hypotheses $H_0: C\beta = d$ can be conducted analogously to the univariate case (Chap. 5) after replacing the log-likelihood, the score function, and the information matrix by their multivariate versions.

Example 6.7 Caesarian Delivery—Hypotheses Testing



We now test whether or not the effects of *NPLAN* and *RISK* differ significantly across both types of infections. Thus we have the hypotheses

$$H_0: \beta_{1N} = \beta_{2N} \text{ and } \beta_{1R} = \beta_{2R}, \quad H_1: \beta_{1N} \neq \beta_{2N} \text{ or } \beta_{1R} \neq \beta_{2R}.$$

This can be written as a special linear hypothesis in the form of $C\beta = 0$, where C has rank 2. As for the Wald test statistic, we have $w = 0.33$ which results in a p-value of 0.8465. Thus H_0 cannot be rejected. Therefore, the two effects do not differ significantly across the two categories.

△

6.5 Bibliographic Notes

Analogously to Sect. 5.6, the concepts of *Bayesian inference* can be transferred to categorical regression models. For *fully Bayesian inference*, we can work either with MH algorithms based on IWLS proposals or with data augmentation schemes. IWLS proposals are described in [Gamerman \(1997\)](#) and [Lenk and DeSarbo \(2000\)](#) for univariate responses. [Brezger and Lang \(2006\)](#) propose a scheme for the multinomial logit model. Sampling schemes for categorical probit models based on data augmentation are more complicated than for binary probit models but similar in nature; see [Albert and Chib \(1993\)](#), [Chen and Dey \(2000\)](#), [Fahrmeir and Lang \(2001\)](#), and [Imai and van Dyk \(2005\)](#). Recently similar sampling schemes have been proposed for multinomial logit models; see [Holmes and Held \(2006\)](#) and [Frühwirth-Schnatter and Frühwirth \(2010\)](#).

Categorical regression models are described in full detail in the books by [Tutz \(2011\)](#) and [Agresti \(2002\)](#). A shorter account can be found in [Fahrmeir and Tutz \(2001, Chap. 3\)](#). The derivation of categorical regression models and more general discrete choice models traces back to McFadden, who was awarded the Noble Prize for Economic Sciences for this work; see, for example, [McFadden \(1984\)](#) or [Train \(2003\)](#).